Computational Modelling of Magnesium Pharmacokinetics

One-Compartment, Two-Compartment, and
Monte Carlo Population Analysis

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GitHub Repository:
https://github.com/BinaryWiz4rd/magnesium-pharmacokinetics

Contents

1 Abstract

This report describes the development and analysis of a computational pharmacokinetic (PK) simulator for magnesium, implemented in Python. Two mathematical models are implemented and compared: a classical one-compartment model and a two-compartment model that captures biphasic plasma kinetics. Four administration routes are simulated: intravenous (IV) bolus, constant-rate IV infusion, oral magnesium citrate, and oral magnesium oxide. A Monte Carlo simulation includes natural differences between patients to show the expected range of magnesium levels across a virtual population. All models are integrated into an interactive Streamlit web dashboard. Simulation results demonstrate that IV infusion is the only route that reliably maintains plasma magnesium within the therapeutic window (0.85–1.10 mmol/L) while avoiding the transient hypermagnesaemia risk of IV bolus. The two-compartment model reveals a substantially higher initial peak for IV bolus than the one-compartment model predicts, owing to the smaller initial distribution volume.

1.1 Physiological role of Magnesium

Magnesium (Mg^{2+}) is the fourth most abundant mineral in the human body and the second most common inside our cells, right after potassium [2, 4]. It acts as an essential helper in 300 to 600 different enzymatic reactions. Specifically, it stabilizes cellular energy production (ATP), aids DNA replication, and supports nerve and muscle communication [3, 4]. Normal blood levels are generally expected to stay between 0.65–1.05 mmol/L [4] or at a specific clinical baseline of 0.85–1.10 mmol/L [1] (and our analysis was taken based on the clinical baseline). Keeping magnesium within this narrow target window is vital during treatment to protect heart and muscle health.

A standard 70 kg adult has about 24 g (1000 to 1120 mmol) of magnesium stored throughout the body [4]. Its location is highly uneven: roughly 50–60% is locked inside the mineral matrix of our bones, while most of the rest sits inside skeletal muscles (∼27%) and soft tissues (∼19%) [4]. Crucially, the fluids outside our cells hold just ∼1% of total body reserves, and only a tiny 0.3% circulates directly in the blood serum [4]. Because a standard blood test only measures this minute circulating fraction, blood levels are a poor indicator of true tissue reserves. This limitation is exactly why we need multi-compartment pharmacokinetic models to map how magnesium actually moves through the whole body.

1.2 Clinical relevance: Hypomagnesaemia

Hypomagnesaemia is one of the most common electrolyte disorders in hospitalized patients, affecting 9–65% of cases depending on the ward's acuity [4]. Clinically, a plasma level below 0.75 mmol/L defines a deficiency [4], while severe drops below 0.50–0.61 mmol/L risk serious complications like cardiac arrhythmias, muscle spasms, seizures, and refractory hypokalemia [4]. Treatment depends on severity:

- **Severe or Symptomatic ($<$ 0.50 mmol/L):** Managed with urgent intravenous magnesium sulfate to quickly refill critical blood and tissue pools [8].
- **Mild to Moderate (0.50–0.75 mmol/L):** Typically treated with oral supplements like highly bioavailable magnesium citrate or magnesium oxide [6, 7].
- **Obstetric Care (Pre-eclampsia/Eclampsia):** Managed via continuous IV infusion to stabilize nerve thresholds and prevent seizures [8].

It is worth mentioning that not all magnesium deficiencies can be detected by a standard blood test. A major diagnostic challenge is latent tetany (spasmophilia) — a condition causing severe muscle and nerve hyper-reactivity despite entirely normal blood magnesium tests [11, 14]. Because blood serum holds a mere 0.3% of the body's total magnesium, a severe deep-tissue deficit can easily hide behind normal blood values [4, 12].

Consequently, diagnosis relies on an electromyography (EMG) ischemic test rather than blood chemistry [14]. By using an inflated arm cuff and hyperventilation, the test triggers temporary localized stress that destabilizes weak nerve membranes. In affected individuals, this causes distinct, rapid nerve firing (doublets or triplets) on the monitor, directly confirming latent tissue-level deficiency [12, 14].

1.3 Pharmacokinetics of Magnesium

1.3.1 Absorption

Oral bioavailability varies depending on the type of magnesium salt used. Magnesium citrate is absorbed well, with a bioavailability around 30–35% ($F \approx 0.31$) and a fast absorption rate ($k_a \approx 0.90$ h^{-1}) [6]. In contrast, magnesium oxide is absorbed poorly ($F \approx 0.04$–0.07, usually modeled as 0.055) and enters the body much slower ($k_a \approx 0.35$ h^{-1}) [7]. Intravenous (IV) magnesium sulfate bypasses the gut entirely, giving it complete bioavailability ($F = 1.0$).

1.3.2 Distribution

After an IV dose, magnesium moves from the bloodstream out into the body's tissues. When using a simple 1-compartment model, the total volume of distribution (V_d) is roughly 0.50 L/kg [8]. Multi-compartment models split this up more realistically: a rapid initial phase fills the bloodstream and highly perfused organs (central volume, $V_1 \approx 0.20$ L/kg), followed by a slower shift into a peripheral compartment ($V_2 \approx 0.30$ L/kg), which represents deeper stores like bone and muscle.

1.3.3 Elimination

The kidneys handle almost all magnesium removal. Although filters screen a massive amount of magnesium daily, the body normally reabsorbs 95–99% of it to prevent wasting [3]. The time it takes to clear half of the drug (half-life, $t_{1/2}$) varies a lot from person to person (10–50 hours) depending on kidney health. For this model, a central half-life of $t_{1/2} = 30$ hours is used, which matches healthy patient baselines in literature [8].

1.3.4 Existing Pharmacokinetic models

Early studies used a simple 1-compartment model. This framework works fine for predicting long-term elimination but fails to capture the sharp, two-part drop seen right after a quick IV injection [9]. While highly complex physiologically based pharmacokinetic (PBPK) models expand this into three- or four-compartment structures to track long-term magnesium exchange and accumulation within bone and deeper tissue reservoirs precisely [13], a standard 2-compartment model offers the best balance for clinical dosing simulations—it stays physiologically accurate without becoming overcomplicated.

1.4 Pathophysiology of Genetic Hypomagnesaemia and Renal Wasting

- **Gitelman Syndrome:** An autosomal recessive salt-losing tubulopathy caused by mutations in the *SLC12A3* gene, which encodes the apical $Na^+ - Cl^-$ cotransporter (NCC) in the distal convoluted tubule (DCT) [15]. Disruption of NCC triggers cellular volume depletion and secondary aldosterone activation, leading to structural DCT atrophy and subsequent downregulation of apical TRPM6 magnesium channels [15]. Clinical manifestations include chronic hypokalemia, significant hypocalciuria, and severe hypomagnesaemia ($C_{\text{base}} \approx 0.55$ mmol/L) driven by a fractional excretion of magnesium (FE_{Mg}) exceeding 4% [15]. Standard lifelong management requires organic formulations like magnesium citrate due to superior bioavailability over inorganic options [15].

- **Bartter Syndrome Type 4:** A severe autosomal recessive disorder caused by mutations in the *BSND* gene encoding barttin, an essential accessory subunit for basolateral chloride channels ($ClC - Ka/ClC - Kb$) in the TAL and DCT [16]. Disruption of these channels abolishes the lumen-positive transepithelial voltage in the TAL, collapsing the primary paracellular driving force responsible for reabsorbing 50%–70% of filtered divalent cations [16]. Phenotypically characterized by antenatal polyhydramnios, salt wasting, and sensorineural deafness, it presents a critically low baseline plasma magnesium ($C_{\text{base}} \approx 0.45$ mmol/L) and an accelerated elimination profile simulated via an intensive scaling factor ($fe_{\text{factor}} = 7.0$) [16].

- **EAST/SeSAME Syndrome:** A complex multi-system disorder caused by loss-of-function mutations in the *KCNJ10* gene encoding the basolateral inwardly rectifying potassium channel Kir4.1 in the DCT [17, 18]. Kir4.1 heteromerizes with Kir5.1 to mediate potassium recycling, which sustains the hyperpolarizing driving force for the active Na^+/K^+-ATPase pump; its disruption depolarizes the basolateral membrane and collapses the electrochemical gradient driving apical magnesium reabsorption [17, 18]. Accompanied by epilepsy, ataxia, and sensorineural deafness, patients exhibit severe renal wasting ($FE_{Mg} \approx 10\%$), anchored at $C_{\text{base}} \approx 0.50$ mmol/L and mathematically replicated using an amplified clearance factor ($fe_{\text{factor}} = 5.0$) [17].

- **CNNM2-Related Hypomagnesaemia:** A rare genetic defect driven by mutations in the *CNNM2* gene encoding cyclin M2, a basolateral magnesium exporter or intracellular sensor localized in the DCT [19]. Pathogenic mutations disrupt protein folding or trafficking, completely blocking transcellular magnesium reabsorption [19]. Because CNNM2 is also expressed in the brain cortex, defects yield a complex neurocognitive spectrum, including intellectual disability and intractable seizures [19]. The renal hallmark is an isolated, profound wasting syndrome with inappropriately elevated fractional excretion (7.5%–12.5%), a critical baseline ($C_{\text{base}} \approx 0.40$ mmol/L), and high therapeutic resistance simulated via an intensive parameter multiplier ($fe_{\text{factor}} = 6.0$) [19].

2 Methods

2.1 Model overview

The simulator implements two PK models sharing the same pharmacological parameters where applicable:

1. **One-compartment model (1C):** drug distributes instantaneously throughout a single homogeneous volume V_d.
2. **Two-compartment model (2C):** drug distributes initially into a central compartment (V_1) and exchanges with a peripheral compartment (V_2) via first-order intercompartmental transfer.

Monte Carlo simulation propagates parameter uncertainty both 1C and 2C models.

2.2 One-Compartment Model

2.2.1 IV Bolus

The plasma concentration at time t following an instantaneous IV dose D administered into volume V_d is:

$$C(t) = C_{\text{base}} + \frac{D}{V_d}\, e^{-k_{el}\, t} \tag{1}$$

where C_{base} is the endogenous baseline concentration and $k_{el} = \ln 2 \,/\, t_{1/2}$ is the first-order elimination rate constant.

2.2.2 IV Constant-Rate Infusion

For a constant infusion rate $R_0 = D/T_{\text{inf}}$ over duration T_{inf}:

$$C(t) = \begin{cases} C_{\text{base}} + C_{ss}\bigl(1 - e^{-k_{el}\, t}\bigr)\,, & 0 \le t \le T_{\text{inf}} \\ C_{\text{base}} + (C_{\text{end}} - C_{\text{base}})\, e^{-k_{el}(t - T_{\text{inf}})}, & t > T_{\text{inf}} \end{cases} \tag{2}$$

where $C_{ss} = R_0 \,/\, (k_{el}\, V_d)$ is the steady-state concentration and $C_{\text{end}} = C(T_{\text{inf}})$ is the concentration at end of infusion.

2.2.3 Oral First-Order Absorption

For formulation with bioavailability F and absorption rate constant k_a:

$$C(t) = C_{\text{base}} + \frac{F\, D\, k_a}{V_d\,(k_a - k_{el})}\,\bigl(e^{-k_{el}\, t} - e^{-k_a\, t}\bigr) \tag{3}$$

This equation assumes $k_a \neq k_{el}$; a numerical guard ($k_a \leftarrow k_a + 10^{-5}$) is applied if the values converge to avoid division by zero.

2.3 Two-Compartment Model

2.3.1 Micro-Rate Constants

Given central volume V_1, peripheral volume V_2, intercompartmental clearance Q, and systemic clearance $CL = k_{el}\, V_d$:

$$k_{10} = \frac{CL}{V_1}, \quad k_{12} = \frac{Q}{V_1}, \quad k_{21} = \frac{Q}{V_2} \tag{4}$$

2.3.2 Hybrid Rate Constants

The system of two coupled ODEs for central and peripheral compartments has closed-form solution characterised by two macro rate constants:

$$\alpha, \beta = \frac{(k_{10} + k_{12} + k_{21}) \pm \sqrt{(k_{10} + k_{12} + k_{21})^2 - 4\, k_{21}\, k_{10}}}{2}, \quad \alpha > \beta \tag{5}$$

The corresponding bolus amplitude coefficients satisfy:

$$A = \frac{\alpha - k_{21}}{\alpha - \beta}, \qquad B = \frac{k_{21} - \beta}{\alpha - \beta}, \qquad A + B = 1 \tag{6}$$

2.3.3 IV Bolus (2C)

$$C_1(t) = C_{\text{base}} + \frac{D}{V_1} \left[A\, e^{-\alpha\, t} + B\, e^{-\beta\, t} \right] \tag{7}$$

2.3.4 IV Infusion (2C)

Using superposition with infusion rate $R_0 = D/T_{\text{inf}}$, define the amplitude terms at end of infusion:

$$q_\alpha = \frac{A}{\alpha} \left(1 - e^{-\alpha\, T_{\text{inf}}}\right), \quad q_\beta = \frac{B}{\beta} \left(1 - e^{-\beta\, T_{\text{inf}}}\right) \tag{8}$$

$$C_1(t) = \begin{cases} C_{\text{base}} + \dfrac{R_0}{V_1} \left[\dfrac{A}{\alpha}(1 - e^{-\alpha t}) + \dfrac{B}{\beta}\left(1 - e^{-\beta t}\right) \right], & t \le T_{\text{inf}} \\ C_{\text{base}} + \dfrac{R_0}{V_1} \left[q_\alpha\, e^{-\alpha (t - T_{\text{inf}})} + q_\beta\, e^{-\beta (t - T_{\text{inf}})} \right], & t > T_{\text{inf}} \end{cases} \tag{9}$$

2.3.5 Oral Absorption (2C)

Via partial-fraction expansion of the Laplace-domain solution for first-order input into a two-compartment system:

$$C_1(t) = C_{\text{base}} + \frac{F\, D\, k_a}{V_1} \left[P\, e^{-\alpha\, t} + Q_c\, e^{-\beta\, t} + R\, e^{-k_a\, t} \right] \tag{10}$$

$$P = \frac{k_{21} - \alpha}{(k_a - \alpha)(\beta - \alpha)}, \quad Q_c = \frac{k_{21} - \beta}{(k_a - \beta)(\alpha - \beta)}, \quad R = -(P + Q_c) \tag{11}$$

The three-term form ensures $P + Q_c + R = 0$, which is the required zero-sum condition from partial fractions (mass balance at $t = 0$ and $t \to \infty$).

2.3.6 Renal Parameterisation via Fractional Excretion of Magnesium (FE_{Mg})

To mathematically incorporate the pathophysiology of genetic renal wasting into the 2-compartment model, the systemic clearance (CL) governing the central elimination rate constant (k_{10}) must be explicitly linked to renal performance. In a physiological state, total systemic clearance is the sum of renal and non-renal elimination components:

$$CL = CL_{\text{renal}} + CL_{\text{non-renal}} \tag{12}$$

The renal clearance component is directly proportional to the Fractional Excretion of Magnesium (FE_{Mg}), which represents the percentage of filtered drug excreted in the final urine:

$$FE_{Mg} = \frac{U_{Mg} \times P_{Cr}}{0.7 \times P_{Mg} \times U_{Cr}} \times 100\% \tag{13}$$

where:

- U_{Mg} and P_{Mg} represent the magnesium concentrations in urine and plasma, respectively.
- U_{Cr} and P_{Cr} denote the creatinine concentrations in urine and plasma, utilized as an endogenous anchor to correct for variations in patient hydration and urine volume.
- The constant 0.7 adjusts for the fact that only approximately 70% of plasma magnesium is free (ultrafiltrable) and capable of passing through the glomerular filtration barrier, while the remaining 30% is bound to proteins.

To map this clinical parameter into the deterministic micro-rate constants of the two-compartment system, the simulator defines a baseline metabolic clearance (CL_{base}) for a healthy individual and introduces a dimensionless renal wasting multiplier, termed the fractional excretion factor (fe_{factor}). The pathological acceleration of drug elimination from the central vascular space is thus parameterised as:

$$CL_{\text{disease}} = CL_{\text{base}} \times fe_{\text{factor}} \tag{14}$$

$$k_{10} = \frac{CL_{\text{disease}}}{V_1} \tag{15}$$

Under normal physiological conditions, a healthy individual responding to hypomagnesaemia dynamically downregulates renal clearance ($FE_{Mg} < 2\%$), which corresponds to $fe_{\text{factor}} = 1.0$. Conversely, for the genetic subgroups investigated, the permanent molecular defects prevent tubular conservation, fixing the elimination profile to a constant, non-compensatory state defined by disease-specific scaling thresholds detailed in the model parameters.

2.4 Monte Carlo simulation

To make the simulation realistic, the model takes into account the fact that no two patients are exactly alike. Before tracking each trajectory, the simulator builds a unique profile for every virtual patient by randomly selecting their traits from statistical distributions.

First, body weight (BW) is chosen using a standard bell curve (normal distribution). To keep the simulation clinically realistic, the allowed weights are capped between 40 and 120 kg:

$$BW \sim \mathcal{N}(\mu_{BW}, \sigma_{BW}), \quad BW \in [40, 120] \text{ kg} \tag{16}$$

Other key bodily traits (θ) — specifically drug half-life, volume of distribution, bioavailability, and the absorption rate constant — can never be zero or negative. To make sure that these values stay strictly positive, they are drawn from log-normal distributions based on their typical averages (μ_θ) and their coefficient of variation (CV_θ):

$$\ln\theta \sim \mathcal{N}\left(\ln\mu_\theta - \tfrac{1}{2}\sigma_{\ln}^2,\ \sigma_{\ln}\right), \quad \sigma_{\ln} = \sqrt{\ln(1 + CV_\theta^2)} \tag{17}$$

The simulator plots a complete magnesium-over-time curve for every single patient. Finally, it stacks all of these individual timelines into a large data matrix ($N \times T$) to extract the 5th, 50th (median), and 95th percentiles. This provides a clear look at both the average patient response and the outer boundaries of the population.

The default simulation settings are configured like below:

- **Total virtual patients (N):** 1,000
- **Half-life variation ($CV_{t_{1/2}}$):** 35%
- **Volume of distribution variation (CV_{V_d}):** 20%
- **Bioavailability variation (CV_F):** 25%
- **Absorption rate variation (CV_{k_a}):** 30%

2.5 PK metrics

Three standard metrics are computed for every simulated profile over a 48-hour simulation window:

- $C_{\max}$: maximum plasma concentration (mmol/L).
- $T_{\max}$: time to $C_{\max}$ (h).
- AUC_{0-48}: area under the concentration-time curve estimated by the trapezoidal rule (mmol·h/L).

2.6 Model parameters

Default parameter values are summarised in Tables 1, 2 and 3.

Table 1: Literature constants and patient assumptions

Parameter	Symbol	Value	Source
Baseline plasma Mg	C_{base}	0.85 mmol/L	[3]
Therapeutic window	—	0.85–1.10 mmol/L	[1]
Volume of distribution	V_d	0.50 L/kg	[8]
Mg citrate bioavailability	F_{cit}	0.31	[6]
Mg oxide bioavailability	F_{ox}	0.055	[7]
Citrate abs. rate const.	$k_{a,\text{cit}}$	0.90 h^{-1}	[6]
Oxide abs. rate const.	$k_{a,\text{ox}}$	0.35 h^{-1}	[7]
Elimination half-life	$t_{1/2}$	30 h	[8]
Molecular weight of Mg	M_w	24.305 g/mol	[3]
Body weight (default)	BW	70 kg	assumption
IV dose	D_{IV}	500 mg elem. Mg	assumption
Oral dose	D_{oral}	300 mg elem. Mg	assumption
IV infusion duration	T_{inf}	4 h	assumption

Table 2: Two-compartment model parameters

Parameter	Symbol	Value	Source / Note
Central volume	V_1	0.20 L/kg	assumption (based on [8])
Peripheral volume	V_2	0.30 L/kg	assumption (based on [8])
Intercompart. clearance	Q	0.12 L/h/kg	assumption
Elimination from central	k_{10}	CL/V_1	derived
Central → peripheral	k_{12}	Q/V_1	derived
Peripheral → central	k_{21}	Q/V_2	derived
Fast macro rate const.	α	1.035 h^{-1}	derived
Slow macro rate const.	β	0.022 h^{-1}	derived; $t_{1/2,\beta} \approx 31$ h

A side note: the total volume of distribution ($V_d = 0.50$ L/kg) from literature was split into central (V_1) and peripheral (V_2) compartments to reflect the fast and slow phases of magnesium distribution, and we just assumed those values basing on the full volume of distribution that was 50 L/kg.

Table 3: Genetic subgroups baseline and renal wasting parameters

Patient Group	Baseline Mg (C_{base})	Empirical FE_{Mg}	Scaling Factor (fe_{factor})	Source
Healthy Individual	0.85 mmol/L	$< 2.0\%$	1.0	[2]
Gitelman Syndrome	0.55 mmol/L	$\approx 5.0\%$	2.5	[15]
EAST/SeSAME Syndrome	0.50 mmol/L	$\approx 10.0\%$	5.0	[17, 18]
CNNM2-Related Defects	0.40 mmol/L	$\approx 12.0\%$	6.0	[19]
Bartter Syndrome Type 4	0.45 mmol/L	$\approx 15.0\%$	7.0	[16]

The two-compartment parameters are calibrated so that the β-phase half-life (≈ 31 h) matches the one-compartment terminal half-life, enabling direct comparison between the two models.

2.7 Software Architecture

The project is implemented in Python 3.10+ using NumPy, Matplotlib, and Streamlit. The codebase is organized into the following modular structure:

```
mg_real_analysis/
  core/             constants.py, derived.py
                    (physiological parameters)
  models/           iv.py, oral.py
                    (1C and 2C ODE mathematical solvers)
                    renal_wasting.py
                    (FE_Mg disease logic & fe_factor constants)
  simulation/       monte_carlo.py
                    (Subgroup generator: age, sex)
  metrics/          calculate.py
                    (Cmax, Tmax, AUC computation)
  dashboard.py      (Interactive Streamlit interface with genetic tabs)
```

Inside the `core/` directory, `constants.py` stores all baseline literature values in simple dictionaries for easy updating, `derived.py` automatically calculates secondary parameters like clearance (CL), volumes (V_d, V_1, V_2), and rate constants (α, β).

The actual math runs inside `models/`, where `iv.py` and `oral.py` use fast, vectorized NumPy functions to solve the 1-compartment and 2-compartment equations, and `renal_wasting.py` applies scaling adjustments to simulate genetic kidney diseases based on fractional excretion (FE_{Mg}) levels.

Population-level data is handled by the simulation engine in `simulation/monte_carlo.py`, which generates the virtual patient cohorts. Finally, `metrics/calculate.py` extracts the peak levels ($C_{\max}$, $T_{\max}$) and total drug exposure (AUC_{0-48}), feeding everything into `dashboard.py` to render the interactive Streamlit web interface, multi-compartment charts, and genetic comparison tabs.

3 Results

3.1 One-Compartment simulation

Figures 1 and 2 show the results and concentration timelines for the 1-compartment (1C) model across all four administration routes.

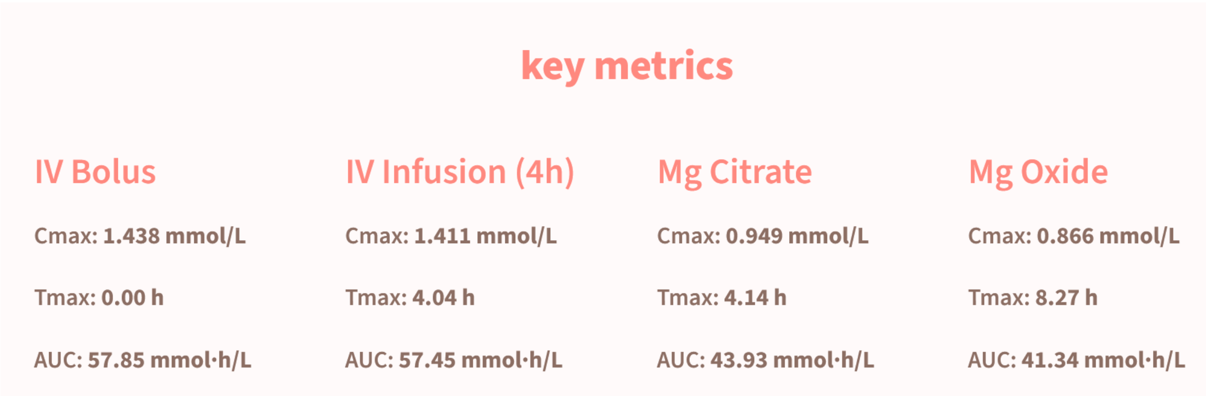


Figure 1: Dashboard panel displaying the calculated 1-compartment pharmacokinetic summary metrics.

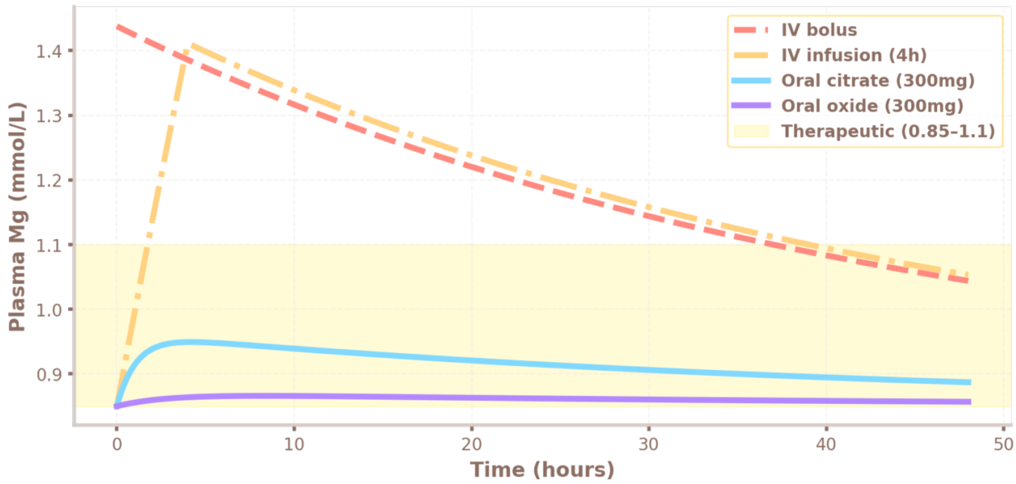


Figure 2: Simulated 1-compartment plasma magnesium concentration-time trajectories across all administration routes against the safe physiological therapeutic window

The dashboard metrics show a clear contrast between injecting magnesium directly into the blood versus taking it orally. The IV Bolus creates an immediate peak concentration ($C_{\max}$) of 1.438 mmol/L at $T_{\max} = 0.00$ hours, shooting well past the safe upper limit of 1.10 mmol/L. Spreading the dose over time using a 4-hour IV Infusion leads to a slightly lower peak of 1.411 mmol/L, which occurs right as the infusion ends ($T_{\max} = 4.04$ h).

For the oral routes, Magnesium Citrate reaches a safer peak of 0.949 mmol/L at $T_{\max} =$ 4.14 hours, staying inside the target window. In contrast, Magnesium Oxide barely changes the baseline due to its poor absorption, peaking at just 0.866 mmol/L after a long delay ($T_{\max} = 8.27$ h). The total exposure to the drug followed such biological rules: the total area under the curve (AUC) is nearly identical for the two IV routes ($\approx$ 57.5 – 57.8 mmol·h/L), while the oral paths show much lower overall exposure (43.93 mmol · h / L for citrate and 41.34 mmol · h / L for oxide) because a large portion of the dose is never absorbed.

3.2 Two-Compartment simulation

Figures 3, 4, and 5 show the quantitative results and curves for the 2-compartment (2C) model.

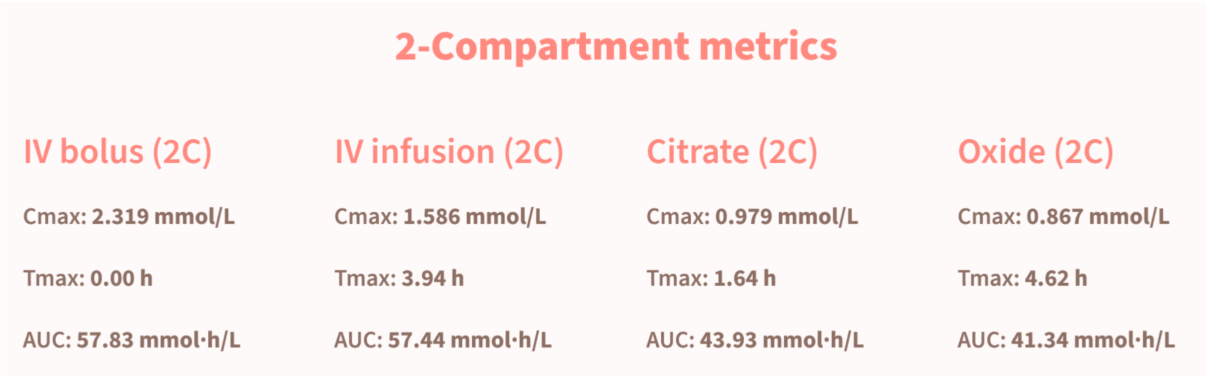


Figure 3: 2C pharmacokinetic metrics from the dashboard screen.

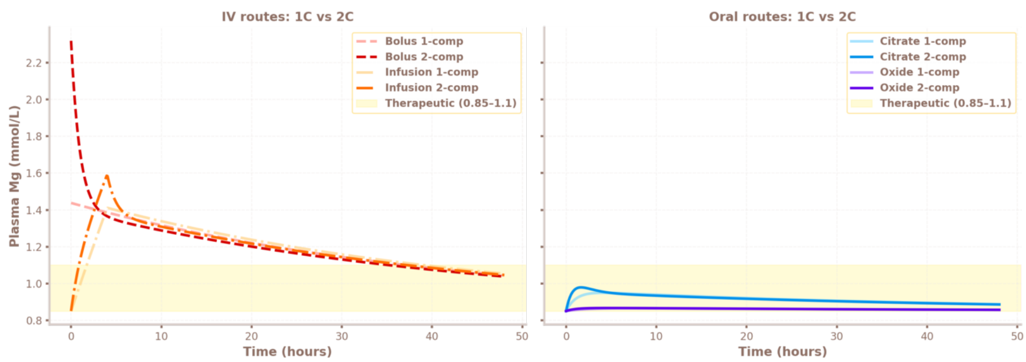


Figure 4: Dashboard interface displaying the 2C concentration-time curves.

Figure 5: Calculated macro-rate distribution constants (α, β) and terminal half-life for the 2C model.

When we compare these metrics to the 1-compartment model, we can see what happens when a dose is locked into a smaller central blood volume ($V_1 = 0.20$ L/kg$\times$70 kg $= 14.0$ L) before it can spread into deeper body tissues. This restriction pushes the IV Bolus peak ($C_{\max}$) up to 2.319 mmol/L and causes Oral Magnesium Citrate to hit its peak faster ($C_{\max} = 0.979$ mmol/L at $T_{\max} = 1.64$ h). On the other hand, the total drug exposure

(AUC_{0-48}) stays similar across all routes (ranging from 41.34 to 57.83 mmol·h/L). This proves that how fast the body cleans out the drug ultimately controls total buildup, regardless of how many compartments are modeled.

The macro-rate constants shown in Figure 5 are responsible for control of this two-step timing. Rapid early spread is driven by $\alpha = 1.0354\ \mathrm{h}^{-1}$, which shows how quickly magnesium moves from the bloodstream into the surrounding tissues. After it is balanced, the system shifts to a slow clearing phase driven by $\beta = 0.0223\ \mathrm{h}^{-1}$ (which gives a long terminal half-life of 31.1 hours). This extended time proves that deeper tissues act as kind of a reservoir, slowly releasing magnesium back into the blood hours later. We discuss these trends and their clinical impact on the treatment of severe muscle spasms (tetany) in Section 4.4 (Concentration-Time profiles).

3.3 Monte Carlo population analysis

To see how natural differences between people affect magnesium levels, we ran a Monte Carlo simulation using 1000 virtual patients. It is possible to change that value in the dashboard. To make sure the random variations are exactly reproducible, we locked the simulator's random seed at 42. We introduced realistic patient variety by varying key bodily traits — specifically drug half-life ($t_{1/2}$) and volume of distribution (V_d)—around the default settings selected in the dashboard control panel. Figure 6 shows the resulting population-wide timelines for a 4-hour intravenous (IV) infusion.

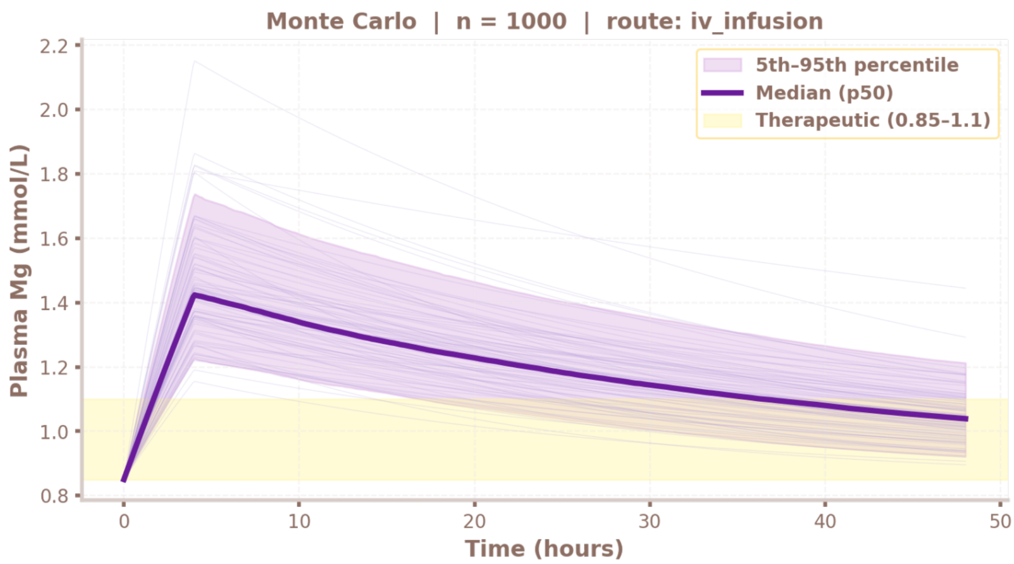


Figure 6: Monte Carlo population simulation for an exemplary IV infusion. The solid bold line represents the population median (p_{50}), while the shaded purple part is the central 90% (5th–95th percentile) population spread. Faint background trajectories are individual virtual patient profiles. The yellow shaded band is the therapeutic target window (0.85–1.10 mmol/L).

As the faint individual "spaghetti" lines in Figure 6 show, differences in body weight and metabolism cause a massive spread in peak magnesium levels and how fast the drug clears. This clearly demonstrates why giving the exact same dose can lead to completely different results from person to person. The shaded purple part highlights where the middle 90% of the simulated population falls, giving us a clear view of the overall clinical risk.

Figure 7: Dashboard calculated pharmacokinetic summary metrics for the 1-compartment Monte Carlo IV infusion simulation

Looking at the dashboard's summary metrics in Figure 7, we can see that there are clear clinical safety concerns. The typical patient response reaches a peak concentration ($C_{\max}$) of 1.424 mmol/L, which is way past the safe upper limit of 1.10 mmol/L. In fact, the 90% population interval for this peak ranges from 1.226 all the way up to 1.738 mmol/L, proving that even patients who clear the drug fastest still face temporary overdosing (hypermagnesemia).

What's more, the simulation shows a median total exposure (AUC) of 57.1 mmol·h/L. Also, the Time in window metric, percentage value means that patients spend a median of just 29.7% of the 48-hour timeline inside the safe therapeutic zone. Because even the lowest group of patients (the 5th percentile) stays above 1.10 mmol/L for nearly 20 hours after the infusion, this standard routine risks keeping almost everyone at toxic levels for hours, rather than just a few isolated cases.

Looking at the dashboard's summary metrics in Figure 7, we can see that there are clear clinical safety concerns. The typical patient response reaches a peak concentration ($C_{\max}$) of 1.424 mmol/L, which is way past the safe upper limit of 1.10 mmol/L. In fact, the 90% population interval for this peak ranges from 1.226 all the way up to 1.738 mmol/L, proving that even patients who clear the drug fastest still face temporary overdosing (hypermagnesemia).

While this specific run evaluates a 1-compartment IV infusion, the dashboard provides the possibility to simulate population variability across all available routes (IV bolus, IV infusion, magnesium citrate, and magnesium oxide) either by the 1-compartment or 2-compartment approach.

3.4 Concentration-Time profiles

In the simulation directly in PyCharm, we generate a dynamic, two-panel visualization comparing the 1-compartment (1C) and 2-compartment (2C) mathematical models side-by-side, shown in Figure 8. This comparison highlights how taking into account peripheral tissue distribution changes predicted serum magnesium levels across different administration routes.

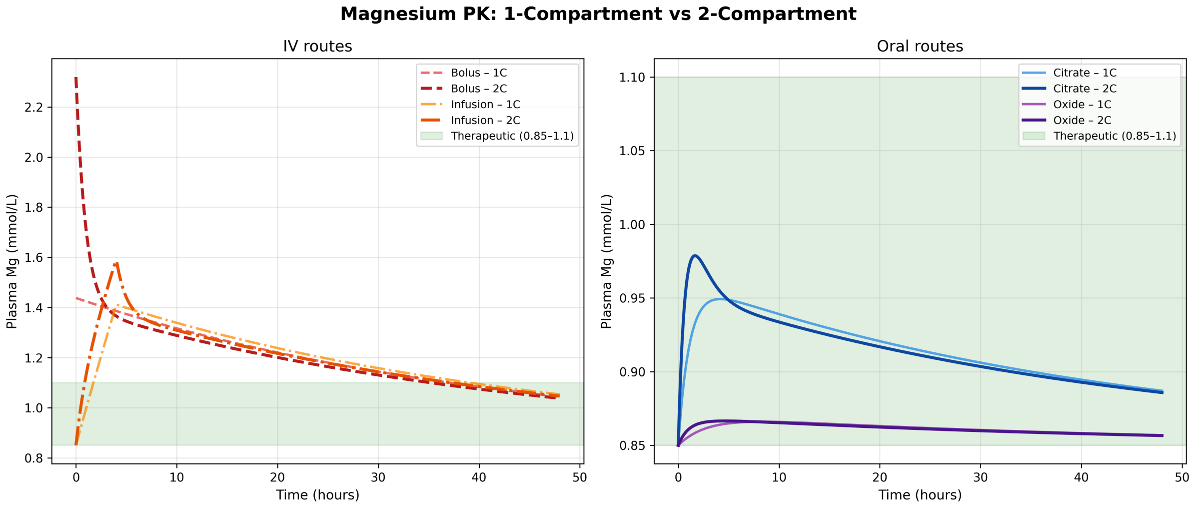


Figure 8: Simulated magnesium concentration-time profiles. Left: IV routes (bolus and infusion), comparing 1C vs. 2C kinetics. Right: oral routes (citrate and oxide) across both models. The green shaded band is the physiological therapeutic window (0.85–1.10 mmol/L).

The left side of Figure 8 shows what happens with the intravenous (IV) routes. For the IV Bolus, the 1C model assumes the magnesium spreads instantly across the entire body, leading to a starting peak concentration (C_{max}) of ~1.44 mmol/L. On the other hand, the 2C model keeps the initial dose locked in the smaller central compartment (V_1) first, pushing that immediate peak (C_{max}) much higher to ~2.32 mmol/L. After this sharp spike, magnesium quickly moves into deeper tissues (V_2) during a rapid balancing phase (called the α-phase), before settling into the slower, steady clearing phase (the β-phase). For the 4-hour IV Infusion, the 2C model shows a clear two-part curve that peaks sharply right when the drip ends before spreading out, while the 1C model tracks a much smoother, classic climb to a steady plateau. Both IV models predict magnesium levels that "shoot" well past the safe upper limit (1.10 mmol/L), highlighting the real-world danger of a temporary overdose (hypermagnesemia) when giving IV magnesium too quickly. But it is worth mentioning that sometimes such application is extremely important, for example in tetany, where the magnesium levels might be a lot lower than the therapeutic threshold, and in severe tetany attack, such IV infusion saves lives, but the important fact is the post-attack care, where the Mg values dropped, and it is important to take that into account in reconvalescence process, to provide accurate medication methods.

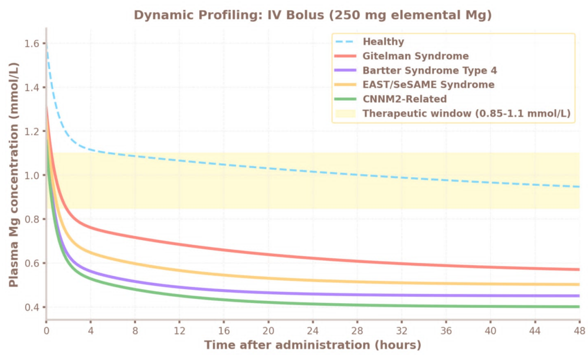
The right side of Figure 8 shows the oral supplement timelines. For Magnesium Citrate ($F = 0.31$), the 2C model predicts a higher, faster peak ($C_{max} \approx 0.98$ mmol/L) than the 1C model ($C_{max} \approx 0.87$ mmol/L). This happens because the absorbed magnesium floods into the tight central compartment before it has a chance to spread out and lower the peak. In total contrast, Magnesium Oxide ($F = 0.055$) produces a nearly flat line that stays right at the body's natural baseline of 0.85 mmol/L in both models. This proves that when a supplement is poorly absorbed, it enters the bloodstream too slowly to cause any sudden shifts. As a result, the mathematical differences between using a 1-compartment or 2-compartment setup completely disappear.

3.5 Simulation of Genetic Disorders

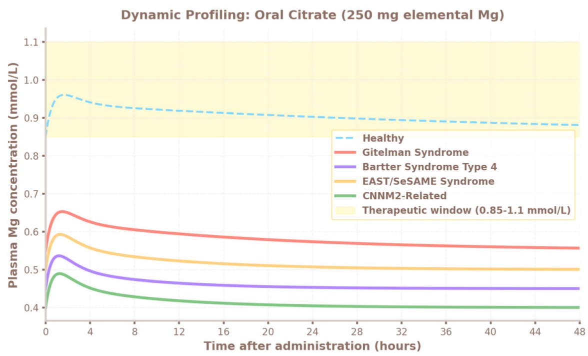
To demonstrate the clinical utility of the developed interactive platform, a dedicated comparative simulation module was constructed to evaluate the dynamic pharmacokinetic shifts across the four distinct hereditary tubulopathy archetypes (Gitelman, Bartter, EAST/SeSAME, and CNNM2-related defects) against a healthy reference baseline ($fe_{\text{factor}} = 1.0$).

The interactive user interface automatically processes the disease-specific adjustments defined in the model parameter tables, applying the chronic baseline depressions (C_{base}) and accelerating the central metabolic elimination constants (k_{10}) via their discrete fractional excretion factors (fe_{factor}). A core strength of the dashboard is its fully reactive, real-time computational engine. By manipulating the sidebar parameters—including the selection of the administration route, adjustments to patient weight, tuning the elemental magnesium dose, or changing the IV infusion duration—the platform instantaneously resolves the underlying pharmacokinetic systems of differential equations. This yields immediate visual updates of the concentration-time curves and automatically re-calculates the quantitative PK metrics in real time.

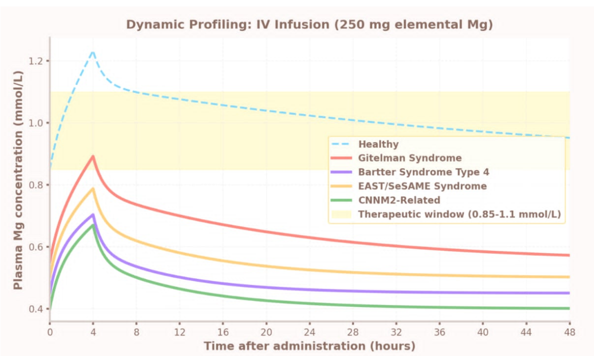
The comprehensive layout of the simulated genetic module and its multi-route evaluation panel is illustrated across Figures 9.



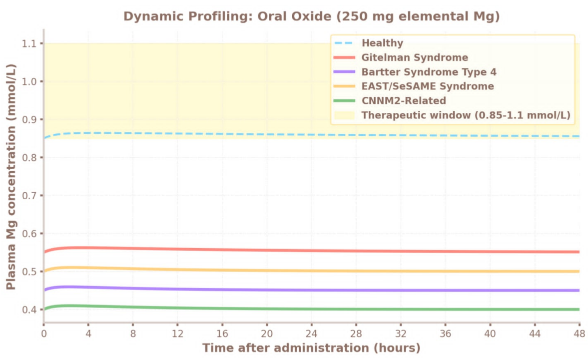
(a) IV Infusion simulation panel.



(b) IV Bolus simulation panel.



(c) Oral Citrate simulation panel.



(d) Oral Oxide simulation panel.

Figure 9: Pharmacokinetic simulation matrix across all four modeled administration routes for healthy and genetic cohorts. Each panel highlights the real-time parameter adaptation and subsequent curve divergence based on disease severity.

Quantitative multi-curve evaluations generated by the dashboard reveal critical insights into route-dependent therapeutic efficacy:

- **Intravenous Routes (Bolus and Infusion):** For all pathologically impacted cohorts, the simulated central concentration curves $C_1(t)$ display a markedly accelerated decay slope. In Bartter syndrome type 4 ($fe_{\text{factor}} = 7.0$), the return to the critically low endogenous baseline occurs rapidly, demonstrating that while intravenous repletion successfully breaches the lower therapeutic margin, the effect is transient due to uncompromised renal wasting.
- **Oral Routes (Citrate and Oxide Salts):** Under a standardized maintenance oral dose of 300 mg, the simulator exposes profound therapeutic resistance in the chronic disease strata. Only oral magnesium citrate successfully elevates the healthy reference curve within the 0.85–1.10 mmol/L therapeutic window. In stark contrast, the profiles for CNNM2-related defects and Bartter syndrome barely shift from their depressed baselines (0.40 mmol/L and 0.45 mmol/L, respectively) and rapidly cascade downward, visually confirming the clinical consensus that standard oral supplementation fails to compensate for high-velocity renal excretion mechanisms.

4 Sensitivity and parameter influence analysis

4.1 Effect of body weight

Body weight directly scales the volume of distribution across both models (V_d in the 1C model; V_1 and V_2 in the 2C model). For instance, increasing a patient's weight from 60 to 90 kg adds more body space for the drug, which drops the IV Bolus peak ($C_{\max}$) by about 33% in both frameworks. On the other hand, total drug exposure (AUC) is completely unaffected by weight. Because we assume that a heavier patient's clearance (CL) scales up perfectly with their size while their elimination rate stays constant, the larger body volume and faster clearing balance each other out.

4.2 Effect of elimination half-life

Drug half-life is the main factor controlling long-term blood levels and total exposure (AUC). For example, doubling $t_{1/2}$ from 30 to 60 hours — which typically happens with severe kidney damage — doubles the total exposure (AUC) but leaves the initial peak ($C_{\max}$) and its timing ($T_{\max}$) virtually identical for IV routes. For oral dosing, a longer half-life pushes the peak ($T_{\max}$) to a later time and keeps blood levels elevated far longer, significantly raising the risk of the drug building up to toxic levels with repeated daily doses.

The Monte Carlo simulation results in Section 4.3 confirm this, showing that the 35% variation in half-life ($CV_{t_{1/2}}$) is the primary reason why the patient uncertainty band widens so dramatically over time, especially during the later hours of the timeline.

4.3 Effect of infusion duration

Slowing down how fast an IV runs directly lowers the highest magnesium spike in the blood. This happens because extending the infusion time spreads the exact same dose over a longer window, giving the body more time to clear the drug as it enters. For a standard 500 mg dose, our model shows that a rapid 1-hour infusion drives the peak concentration ($C_{\max}$) up to a dangerous 1.77 mmol/L, well above the safe upper limit of 1.10 mmol/L.

This reveals a vital safety rule for clinicians: simply stretching the infusion to 4 hours is not enough, as the typical patient still overshoots the window at 1.41 mmol/L. In fact, even a very slow 12-hour drip only manages to bring the peak down to a borderline level of 1.13 mmol/L. These results prove that adjusting the infusion duration alone cannot guarantee patient safety; to keep levels within the therapeutic window, doctors must either lower the total dose or actively track blood levels with therapeutic drug monitoring.

4.4 Effect of Two-Compartment Volume Parameters

In the 2-compartment model, the initial peak concentration ($C_{\max}$) is heavily dictated by the size of the central blood volume (V_1) and how fast the drug leaves it. Shrinking V_1 (e.g., from 0.20 to 0.12 L/kg) means the same magnesium dose is trapped in a smaller initial fluid space, driving the immediate $C_{\max}$ spike much higher without altering the total long-term exposure (AUC). Conversely, increasing the intercompartmental clearance (Q) speeds up this redistribution — it drops the initial blood peak by pulling the drug out of the central circulation and into the deeper tissue compartment (V_2) much faster.

To make these relationships intuitive, these parameters can be adjusted interactively via the dashboard sidebar, allowing users to visually track how different tissue distribution speeds alter the patient's immediate drug profile.

4.5 Bioavailability sensitivity (oral routes)

The oral delivery paths show a near-linear relationship between bioavailability (F) and both the peak concentration ($C_{\max}$) and total exposure (AUC). For example, doubling the citrate bioavailability from 0.31 to 0.62 doubles its AUC and pushes its peak ($C_{\max}$) from a safe 0.95 mmol/L up to 1.38 mmol/L. Meanwhile, the magnesium oxide formulation is so severely restricted by its poor absorption that even if it reaches its highest biologically plausible limit ($F = 0.10$), its total exposure (AUC) still lags far behind the IV routes.

4.6 Effect of Genetic Scaling Factors

The dimensionless fractional excretion multiplier (fe_{factor}) heavily dictates systemic drug exposure (AUC) and clearance velocity from the central vascular space. Because systemic clearance scales linearly with this parameter ($CL_{\text{disease}} = CL_{\text{base}} \times fe_{\text{factor}}$), increasing fe_{factor} from 1.0 up to 7.0 induces a strict, inversely proportional reduction in AUC_{0-48} for any given dose.

Furthermore, while a high fe_{factor} does not mechanically alter the initial post-bolus $C_{\max}$, it drastically accelerates the decay slope of both the distribution and elimination phases [15, 17]. This mathematical acceleration explains the rapid collapse of simulated concentration curves back to sub-physiological, depressed baselines, validating clinical observations that standard oral maintenance doses are structurally insufficient to stabilize plasma levels in severe genetic wasting phenotypes [16, 19].

5 Discussion and Conclusions

5.1 Summary of findings

1. **IV Bolus carries the highest risk:** Both models predict a peak (C_{max}) well above the safe therapeutic window (1.44 mmol/L in 1C vs. 2.32 mmol/L in 2C for a 500 mg dose). The 2C model reveals that the simpler 1C framework drastically underestimates this dangerous, immediate plasma spike.

2. **IV Infusion remains the standard for acute care:** Although a 500 mg dose over 4 hours still pushes the peak above 1.10 mmol/L, the gradual delivery gives the body time to buffer and redistribute the mineral. Slight tweaks to the dose or drip rate can easily keep the patient safely within the target zone.

3. **Oral Citrate is preferred for non-urgent use:** It delivers a safe peak of 0.95 mmol/L, staying comfortably within the therapeutic window. Our Monte Carlo analysis confirms its reliability, showing that over 90% of the simulated population achieves a predictable peak between 0.86 and 1.03 mmol/L.

4. **Oral Oxide is inadequate for rapid correction:** Due to its severely limited bioavailability, the resulting peak concentration barely shifts the patient's baseline, making it useless for treating urgent deficiencies regardless of the dose size.

5. **Renal function is the critical covariate.** Monte Carlo results show that half-life variability ($CV = 35\%$) is the dominant source of population uncertainty; patients with impaired renal clearance face accumulation risk with any route.

6. **Genetic tubulopathies cause significant drug resistance.** The simulations based on disease parameters (fe_{factor}) reveal that conventional oral administration does not result in achieving drug levels in the desired range of 0.85-1.10 mmol/L owing to unaccounted renal losses [15, 19]. Even the intravenous bolus dose provides just a temporary peak level, after which the profile falls to depressed baseline levels ($C_{\text{base}} \in [0.40, 0.55]$ mmol/L) [16, 17]. This highlights the importance of continuous infusion over conventional dosing in this genetic group, aligning with clinical observations of profound therapeutic resistance [15].

5.2 Model Limitations

- **Fixed baseline:** $C_{\text{base}} = 0.85$ mmol/L is assumed constant in the general model; in clinically hypomagnesaemic genetic cohorts, the true baseline is lower, and dynamic endogenous renal regulation is not fully modeled.

- **Linear Kinetics and Homeostasis:** All models assume linear, first-order kinetics. This omits dynamic feedback loops and saturable renal transport pathways—such as the maximum tubular transport capacity (T_m)—that govern physical magnesium reabsorption in response to fluctuating plasma concentrations [3, 5].

- **Bone exchange compartment:** The two-compartment structure approximates peripheral distribution as a single lumped space, omitting the distinctly slow, deep exchange with the mineralised bone matrix that dominates at long time scales (> 48 h) [4].

- **Parameter Shift and Pathology:** Standard baseline pharmacokinetic variables were sourced from small studies on healthy controls. Consequently, they fail to capture complex systemic metabolic alterations triggered by severe, multi-system genetic pathologies, such as the extrarenal complications inherent to EAST/SeSAME syndrome [17].
- **Omission of protein binding:** Plasma magnesium protein binding ($\sim 30\%$) and potential drug-drug interactions were neglected. Treating the entire plasma pool as ultrafiltrable represents a localized simplification of systemic clearable mass fractions [2].

5.3 Future directions

- **Three-compartment model** including bone mineralisation compartment for long-term kinetic predictions.
- **Renal covariate model** incorporating estimated GFR as a continuous covariate on k_{el} to personalise predictions for patients with renal impairment.
- **Population PK (NLME) fitting** to real clinical plasma concentration data (the `mg_real_analysis/` module provides the intended framework).
- **Repeated dosing simulations** to model accumulation and time to steady state.
- **Bayesian forecasting** — using the Monte Carlo framework as a prior and updating with individual patient observations for dose individualisation.
- **Mechanistic Non-linear Kidney Sub-modelling:** Upgrading the static fe_{factor} to a mechanistic sub-model utilizing Michaelis-Menten kinetics. This would simulate active transcellular (TRPM6) and passive paracellular transport, capturing how specific mutations dynamically lower the transport maximum (T_m) under varying filtered loads [3, 18].
- **Dynamic Dosing Optimizers:** Leveraging Streamlit to build a backward-solving therapeutic optimizer. This module would automatically calculate multi-dose oral schedules or continuous micro-infusion rates needed to stabilize plasma levels within the target window, mitigating the rapid elimination seen in severe CNNM2 or Bartter phenotypes [15, 19].
- **Implementation of Patient Subgroup Stratification:** Expanding the population simulator to account for age, sex, and specific physiological states such as pregnancy. This would involve dynamically scaling the baseline GFR and expanding the central distribution volume V_1 by 20–30% to capture gestational plasma volume expansion, which is critical for treating vulnerable patient cohorts.

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